

Curcumin in inflammatory diseases.



Curcumin (diferuloylmethane), a yellow coloring agent extracted from turmeric is also used as a remedy for the treatment and prevention of inflammatory diseases. Acute and chronic inflammation is a major factor in the progression of obesity, type II diabetes, arthritis, pancreatitis, cardiovascular, neurodegenerative and metabolic diseases, as well as certain types of cancer. Turmeric has a long history of use in Ayurvedic medicine for the treatment of inflammatory disorders. Recent studies on the efficacy and therapeutic applicability of turmeric have suggested that the active ingredient of tumeric is curcumin. Further, compelling evidence has shown that curcumin has the ability to inhibit inflammatory cell proliferation, invasion, and angiogenesis through multiple molecular targets and mechanisms of action. Curcumin is safe, non-toxic, and mediates its anti-inflammatory effects through the down-regulation of inflammatory transcription factors, cytokines, redox status, protein kinases, and enzymes that all promote inflammation. In addition, curcumin induces apoptosis through mitochondrial and receptor-mediated pathways, as well as activation of caspase cascades. In the current study, the anti-inflammatory effects of curcumin were evaluated relative to various chronic inflammatory diseases. Based on the available pharmacological data obtained from in vitro and in vivo research, as well as clinical trials, an opportunity exists to translate curcumin into clinics for the prevention of inflammatory diseases in the near future. Copyright © 2012 International Union of Biochemistry and Molecular Biology, Inc.